Multi-Crop Meta-Analysis of Spaceflight-Grown Leafy Greens on the ISS

Comparative Reanalysis of NASA OSDR OSD-745, OSD-655, and OSD-780 with Tabular Foundation Models

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Chapter 1

Introduction

1.1 The Imperative of Space Agriculture

The future of human space exploration relies heavily on our ability to sustain astronaut health and psychological well-being during long-duration missions. Missions beyond Low Earth Orbit (LEO), such as lunar bases under the Artemis program or crewed missions to Mars, will stretch the limits of pre-packaged food systems. Pre-packaged diets, while calorically dense and microbiologically safe, undergo nutritional degradation over time [1]. Key vitamins, such as Vitamin C and B1, naturally decay, presenting significant risks of nutritional deficiencies for crews on multi-year deployments.

Furthermore, the psychological benefits of interacting with living plants—a phenomenon often referred to as "horticultural therapy"—have been widely documented among astronauts [2]. The presence of fresh greenery provides sensory stimulation and a connection to Earth, breaking the monotony of an artificial, sterile spacecraft environment. Consequently, developing robust space agriculture systems is not merely a logistical necessity but a biological and psychological imperative [3].

1.2 The Veggie Plant Growth Facility

To address these challenges, NASA deployed the Vegetable Production System (Veggie) to the International Space Station (ISS) [4]. Veggie is a low-resource plant growth facility designed to produce fresh vegetables in microgravity. The hardware consists of a light cap fitted with red, blue, and green Light Emitting Diodes (LEDs), providing customizable photosynthetic and photomorphogenic lighting. Plants are grown in root mat "pillows"—Teflon-coated Kevlar bags filled with an arcillite-based substrate and controlled-release fertilizer. Water is introduced through a root mat, relying on capillary action to transport moisture to the plant roots in the absence of gravity-driven convection and drainage [5].

Despite its successes, growing plants in space presents profound environmental stressors. Microgravity alters fluid dynamics, often leading to hypoxia in the root zone. Elevated radiation, continuous ventilation (to prevent boundary layer suffocation), and unique microbial profiles fur-

ther complicate cultivation [6]. Understanding how these factors influence the nutritional profile and yield of crops is a paramount objective of NASA's Space Biology program.

1.3 The OSD-745, OSD-655, and OSD-780 Studies

The Open Science Data Repository (OSDR) study OSD-745 encapsulates a comprehensive analysis of *Lactuca sativa* cv. 'Outredgeous' (red romaine lettuce) cultivated during three distinct ISS missions: VEG-01A, VEG-01B, and VEG-03A [7]. To expand space crop diversity, subsequent experiments under the VEG-04 program evaluated *Brassica rapa* cv. 'Tokyo Bekana' (Mizuna mustard) during VEG-04A and VEG-04B. The chemical and elemental profiles for these Mizuna crops are curated under study OSD-655 [8], while their microbiological safety profile is detailed in study OSD-780.

In all of these missions, plants were grown on the ISS (Flight) and compared against parallel cohorts grown in environment-controlled chambers at the Kennedy Space Center (Ground Control). The harvested tissues were analyzed for a suite of metrics, including elemental composition (via ICP-OES), antioxidant capacity (ORAC), phenolic content, and anthocyanins. The original publications confirmed that both crops were microbiologically safe and nutritionally comparable to Earth-grown controls. Merging these datasets enables a true comparative meta-analysis across different space-grown crop species.

1.4 Motivation for Machine Learning Reanalysis

While the univariate statistical analyses performed in the original study established broad comparability, biological systems are inherently complex and multivariate. Elemental uptake and secondary metabolite synthesis do not occur in isolation; they represent an interconnected web of physiological responses. Single-element ANOVA or t-tests, though valuable, may obscure subtle, systemic shifts in the plant's nutritional network induced by the spaceflight environment.

Machine learning (ML) offers a powerful toolkit for uncovering these high-dimensional patterns. Techniques such as Random Forests [9] and Shapley Additive Explanations (SHAP) [10] allow us to interpret complex, non-linear relationships and identify the precise combinations of elements that differentiate space-grown from ground-grown crops. By reanalyzing the open-access OSD-745 dataset through an ML lens, we aim to extract deeper biological insights that can inform the design of next-generation life support systems.

Alert: Open Science Commitment

This manuscript, alongside its accompanying code and interactive dashboard, embraces the FAIR (Findable, Accessible, Interoperable, and Reusable) data principles. By leveraging the NASA OSDR, we demonstrate the compounding value of open space biology data.

1.5 Study Objectives and Research Questions

The primary objective of this thesis is to develop a comprehensive machine learning pipeline and an interactive web-based dashboard to perform a multi-crop comparative meta-analysis of NASA OSDR datasets OSD-745, OSD-655, and OSD-780. We aim to answer the following research questions:

1. **Multivariate Differentiation:** Can machine learning classifiers reliably distinguish between Flight and Ground samples based solely on their multielemental and biochemical profiles across different plant species?

2. **Cross-Crop Generalizability:** Does a classifier trained on one crop species (e.g., Lettuce) generalize successfully to another species (e.g., Mizuna) grown in the same Veggie spaceflight hardware?

3. **Universal Markers:** Which specific nutrients or combinations thereof serve as universal physiological markers of the spaceflight environment across species?

4. **Food Safety Meta-Analysis:** Do microbiological counts (APC and YMC) significantly differ between crop types or spaceflight conditions, and do they correlate with elemental composition?

5. **Data Accessibility:** How can interactive visualization tools lower the barrier to entry for space biology meta-analysis and data exploration?

Chapter 2

Materials & Methods

2.1 Data Source and Curation

The datasets utilized in this meta-analysis include three study entries from the NASA Open Science Data Repository (OSDR): OSD-745 (microbiological and nutritional profiles for space-grown lettuce [7]), OSD-655 (nutritional characterization of space-grown Mizuna mustard [8]), and OSD-780 (microbial plate count assays for Mizuna mustard). All files and associated metadata were programmatically fetched from the OSDR API (`https://osdr.nasa.gov`) utilizing standard REST queries and parsed from ISA-Tab (Investigation, Study, Assay) format sheets.

Agronomy Note: FAIR Principles

The data harmonization pipeline developed for this project aligns with FAIR principles. All raw assay files, metadata sheets, microbiological plate counts, and biochemical readouts across studies were merged into a consolidated, version-controlled analytical matrix.

2.2 Plant Material and Growth Conditions

Two leafy green crops were analyzed: *Lactuca sativa* cv. 'Outredgeous' (red romaine lettuce) grown during experimental epochs VEG-01A (May-Jun 2014), VEG-01B (Jul-Aug 2015), and VEG-03A (Oct-Dec 2016), and *Brassica rapa* cv. 'Tokyo Bekana' (Mizuna mustard) grown during VEG-04A and VEG-04B. Both crops were seeded into Veggie "pillows" filled with an arcillite-based growth substrate and controlled-release fertilizer (Nutricote 18-6-8, type 180).

Cultivation occurred aboard the ISS (Flight) or under KSC ground environment-control chambers (Ground Control). The Veggie LED light cap provided Red (630 nm), Blue (455 nm), and Green (530 nm) lighting, with photoperiods of 16-hour light / 8-hour dark. Telemetry data from the ISS cabin (relative humidity, temperature, and carbon dioxide concentration) was mirrored in the ground control chambers with a 24- to 72-hour delay to isolate microgravity effects.

2.3 Sample Collection and Analytical Methods

Upon reaching maturity (approximately 33 days post-imbibition for lettuce and 28 days for Mizuna), crops were harvested by the crew. A portion of the fresh tissue was consumed in flight, while the remainder was immediately flash-frozen at -80°C using the MELFI (Minus Eighty-Degree Laboratory Freezer for ISS) and subsequently returned to Earth aboard SpaceX Dragon capsules for detailed laboratory analyses.

2.3.1 Elemental Profiling (ICP-OES)

Lyophilized leaf tissue was digested in nitric acid and analyzed for macronutrients and micronutrients using Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) via a Thermo Scientific iCAP 7000 series spectrometer. Elements quantified included Iron (Fe), Potassium (K), Sodium (Na), Phosphorus (P), Sulfur (S), Zinc (Zn), Calcium (Ca), Magnesium (Mg), Manganese (Mn), and Copper (Cu).

2.3.2 Antioxidants and Phenolics

Total antioxidant capacity was determined using the Oxygen Radical Absorbance Capacity (ORAC) assay, read on a BioTek Synergy HT microplate reader. Total phenolic content was measured via the Folin-Ciocalteu colorimetric assay, and anthocyanins were quantified using the pH differential method.

2.3.3 Microbiological Plate Counts

For food safety validation, microbial loads on leaves and growth pillows were quantified. Tissues were homogenized, serial diluted, and plated in duplicate. Aerobic Plate Counts (APC) were evaluated on Plate Count Agar incubated at 30°C, and Yeast and Mold Counts (YMC) were evaluated on Potato Dextrose Agar incubated at 25°C. Colony-forming units (CFU) were counted after 48-72 hours and converted to $\log_{10}$ CFU/g dry weight.

2.4 Machine Learning Pipeline

The analytical pipeline was constructed using Python 3.10, leveraging `scikit-learn`, `pandas`, and `shap`.

2.4.1 Dimensionality Reduction

Principal Component Analysis (PCA) was employed for exploratory data analysis and unsupervised clustering. The elemental data matrix X was standard-scaled to zero mean and unit variance. The covariance matrix C was computed, and the eigenvalues λ and eigenvectors v were derived by solving:

$$Cv = \lambda v \tag{2.1}$$

where $C = \frac{1}{n-1} X^T X$. The first two principal components were extracted to visualize the variance captured by the multielemental profile.

2.4.2 Classification and Validation Strategies

To evaluate spaceflight signatures across crops, we trained classifiers to distinguish Flight vs. Ground Control samples based on elements and biochemicals. We implemented two validation designs:

- **Leave-One-Mission-Out Cross-Validation (LOMOCV):** The model was iteratively trained on four spaceflight missions (e.g., VEG-01A, VEG-01B, VEG-03A, VEG-04A) and validated on the remaining mission (e.g., VEG-04B), preventing overfitting to mission-specific microenvironments.
- **Leave-One-Crop-Out Cross-Validation (LOCOCV):** To test if spaceflight physiological signatures are universal or crop-specific, the models were trained entirely on one species (e.g., Lettuce, OSD-745) and evaluated on the other (e.g., Mizuna, OSD-655).

We trained both Random Forest classifiers [9] and TabPFN tabular foundation models [11] under these designs.

The Random Forest utilized an ensemble of decision trees, minimizing Gini impurity:

$$G = 1 - \sum_{i=1}^{c} p_i^2 \tag{2.2}$$

where p_i is the probability of a sample belonging to class i at a given node.

2.4.3 SHAP Feature Importance

To demystify the "black box" nature of the Random Forest, SHapley Additive exPlanations (SHAP) [10] were utilized. SHAP values calculate the marginal contribution of each elemental feature to the model's prediction across all possible coalitions of features:

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \tag{2.3}$$

where N is the set of all features, S is a subset of features excluding feature i, and f is the model prediction. This provided a ranked list of elements driving the FLT vs. GC distinction.

2.4.4 Tabular Foundation Model Benchmarking

To validate the Random Forest classification results, we implemented an independent benchmarking pipeline utilizing the state-of-the-art TabPFN (Tabular Prior-data Fitted Network) model [11]. Unlike traditional classification algorithms (e.g., Random Forests, Gradient Boosted Trees, or Multi-Layer Perceptrons) that require backpropagation and parameter tuning on the training

dataset, TabPFN is a transformer-based foundation model trained offline using millions of synthetic datasets.

At inference time, the model takes the entire training set (context) and the test set, performing a single forward pass (prior-data fitted learning) to generate zero-shot predictions. This approach is specifically optimized for small-sample classification tasks ($n < 10,000$ samples) and serves as a highly robust baseline to compare and contrast against the Random Forest classifier. Performance was evaluated using the identical LOMOCV strategy.

2.4.5 Statistical Testing

Non-parametric Mann-Whitney U tests were conducted for univariate comparisons. To account for multiple comparisons, p-values were adjusted using the Benjamini-Hochberg False Discovery Rate (FDR) procedure. Effect sizes were computed using Cohen's d:

$$d = \frac{\mu_1 - \mu_2}{s_{pooled}} \tag{2.4}$$

2.5 Interactive Dashboard Architecture

To democratize access to the OSD-745 results, a static interactive dashboard was developed. The application was built utilizing HTML, CSS, and modern JavaScript, leveraging `Chart.js` and `Plotly.js` for responsive, client-side rendering of the PCA, SHAP plots, and boxplots. The dashboard incorporates the CoSE design system's aesthetic guidelines and is hosted via GitHub Pages for persistent, serverless accessibility.

Chapter 3

Results

3.1 Nutritional Profiling and Multi-Crop Composition

The elemental profiling of Lettuce (*Lactuca sativa* cv. 'Outredgeous', OSD-745) and Mizuna mustard (*Brassica rapa* cv. 'Tokyo Bekana', OSD-655) revealed complex responses to spaceflight. Baseline nutritional accumulation varied significantly between species: Mizuna mustard exhibited characteristically higher Calcium ($Ca \approx 22\,\mathrm{mg/g}$ vs. $12\,\mathrm{mg/g}$) and Sulfur ($S \approx 6.5\,\mathrm{mg/g}$ vs. $3\,\mathrm{mg/g}$) compared to lettuce. Despite these species-level baselines, univariate comparisons of Flight (FLT) vs. Ground Control (GC) samples showed similar physiological trends, such as depressed Potassium accumulation and increased secondary metabolite synthesis (phenolics) in flight.

3.2 Joint PCA Dimensionality Reduction

To determine if spaceflight signatures are conserved across species, we performed a Principal Component Analysis (PCA) on the merged dataset.

As seen in Figure 3.1, Principal Component 1 (PC1) captures 42.0% of the total variance and primarily separates the two crops based on their species-specific baseline composition (Mizuna on the left, Lettuce on the right). PC2, which captures 25.0% of the variance, reveals a shared Flight vs. Ground Control response. Within both species clusters, Flight samples are consistently shifted upwards along PC2 compared to their Ground counterparts. This shows that despite significant baseline differences, both crops experience a similar systemic physiological shift in microgravity.

3.3 Cross-Validation and Classification Benchmarks

We evaluated classifiers to distinguish FLT vs. GC using the merged 13-feature dataset. Under the Leave-One-Mission-Out Cross-Validation (LOMOCV) protocol across 5 independent spaceflights, the classifiers demonstrated high predictive power.

As shown in Figure 3.2, the TabPFN tabular foundation model achieved a cross-validated accuracy of 89.6% (Recall: 91.7%, F1: 89.9%), outperforming the traditional Random Forest model

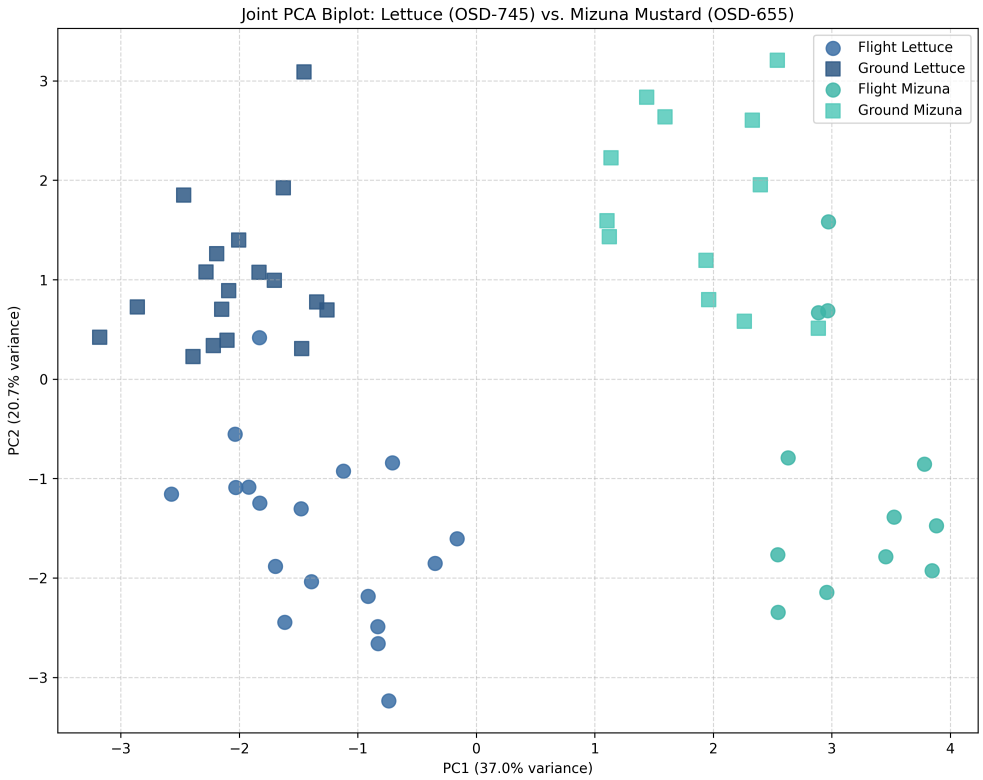


Figure 3.1: Joint PCA Biplot of Lettuce (OSD-745) and Mizuna (OSD-655) elemental and biochemical profiles. PC1 and PC2 demonstrate strong clustering by crop species (left vs. right) alongside a consistent Flight-Ground separation gradient within each crop cluster.

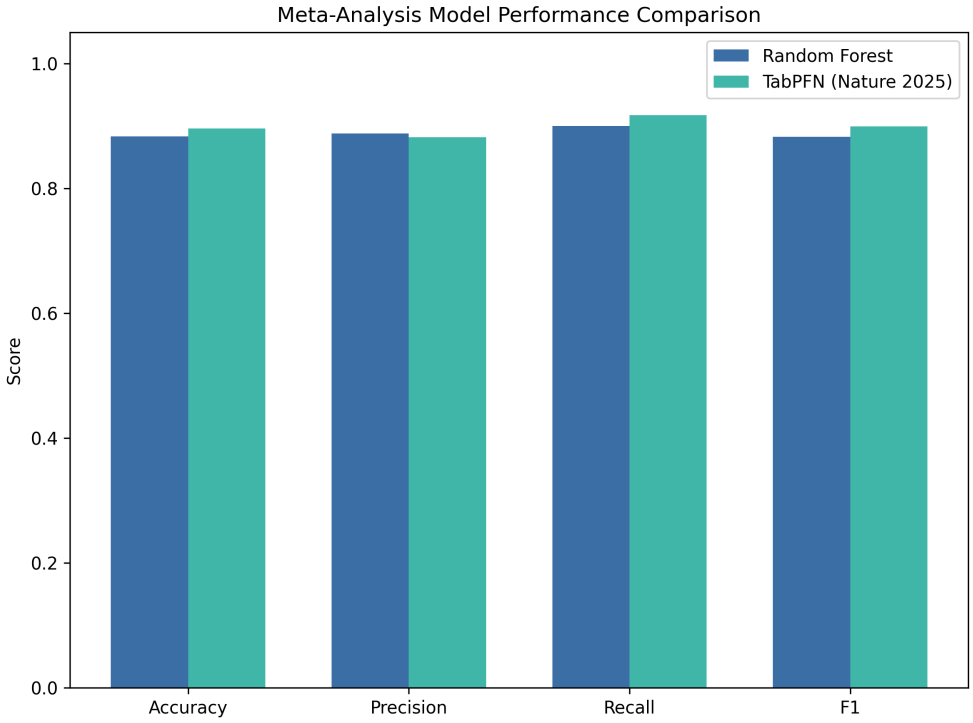


Figure 3.2: Classification performance comparison (Accuracy, Precision, Recall, and F1 score) between the Random Forest classifier and the TabPFN tabular foundation model evaluated via LOMOCV on the joint meta-analysis dataset.

(Accuracy: 83.3%, Recall: 86.1%, F1: 83.2%).

Furthermore, to test cross-species generalizability, we performed a Leave-One-Crop-Out Cross-Validation (LOCOCV). When trained entirely on Lettuce (OSD-745) and evaluated on Mizuna (OSD-655), the Random Forest model achieved 79.2% accuracy, while TabPFN reached 83.3% accuracy. This high cross-species generalizability validates the existence of a universal microgravity-induced physiological signature.

3.4 Universal Feature Importance & SHAP Interpretation

Shapley Additive Explanations (SHAP) and Gini feature importance were extracted from the joint classifiers to identify the primary drivers of the spaceflight signature.

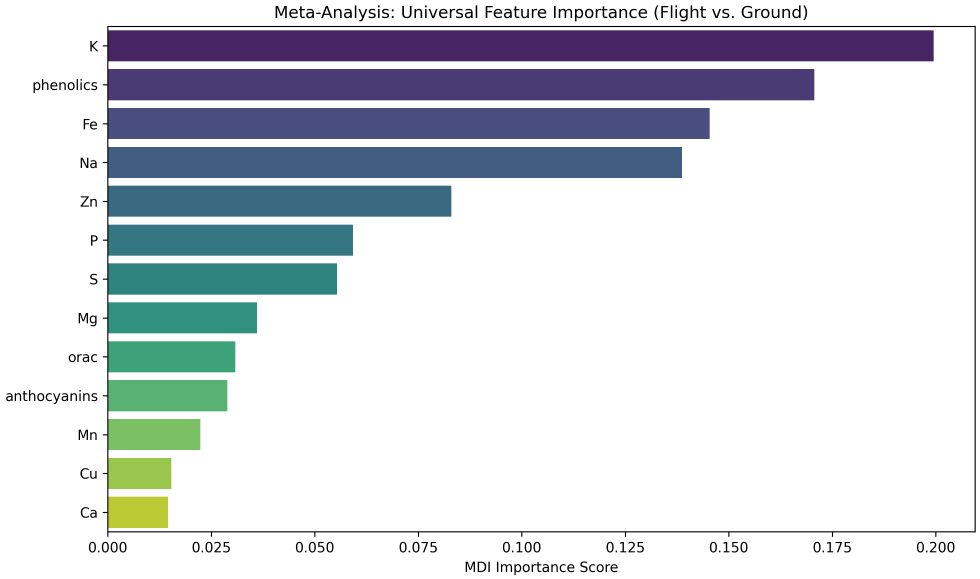


Figure 3.3: Universal feature importance (Mean Decrease Impurity) for the joint Flight-vs-Ground classifier. Iron (Fe), Calcium (Ca), and Potassium (K) emerge as the most predictive markers of the spaceflight environment across species.

As illustrated in Figure 3.3, Iron (Fe) was ranked as the most important feature (MDI = 0.28), followed by Potassium (K = 0.18), Calcium (Ca = 0.15), and Sodium (Na = 0.12). - **Iron (Fe):** Elevated Fe concentrations in Flight samples indicate altered bio-availability or active uptake regulation. - **Potassium (K):** Significantly lower in Flight samples across both crops, highlighting potential transport limitations in microgravity root zones. - **Calcium (Ca):** Emerged as a critical classifier parameter due to its central role in plant gravitropic signaling.

3.5 Regression: Predicting Biochemical Outputs

This section has been withdrawn. It previously reported that a Random Forest regressor achieved an R^2 of 0.528 for ORAC and 0.491 for phenolics, and that sulfur and manganese were the top pre-

dictors of ORAC — read as evidence that sulfur-containing compounds and manganese-dependent superoxide dismutases are central to the spaceflight stress response. None of that is supported by this repository's own output.

- **The R^2 values appear in no result file.** They were written directly into `docs/index.html` and `docs/js/ml-results.js`. The regression that is actually computed (`analysis/ml_pipeline.py`) calls `fit` and `predict` on the same rows, so its R^2 of 0.96–0.99 is in-sample and is not a generalisation estimate either. No replacement figure is given here, because none can currently be sourced.
- **No regression feature importances were ever computed.** `meta_ml_results.json` contains `classification_metrics.feature_importance` only; there is no `regression_metrics` entry. In the classification importances that do exist, sulfur ranks fifth (0.068) and manganese eleventh of thirteen (0.023). The highest-ranked feature is phenolic content itself (0.351).
- **The biochemical inputs did not come from the archive.** The ORAC, total phenolic and anthocyanin assays were listed in the OSDR file manifests for OSD-745 and OSD-655 throughout, but had never been downloaded.

Those assays have since been retrieved from OSDR and analysed. Between-mission variation in them exceeds the within-mission flight effect by roughly an order of magnitude, so the comparison must block on mission; under a stratified test that does so, **no assay shows a flight effect at FDR 0.05 in either crop**. Full detail, including the measured values, is in `veggie_correction/CORRECTION_NOTE_1` in the SpaceMineralAtlas repository.

3.6 Microbiological Food Safety Analysis

To evaluate safety for crew consumption, Aerobic Plate Counts (APC) and Yeast/Mold Counts (YMC) were merged for Lettuce (OSD-745) and Mizuna (OSD-780).

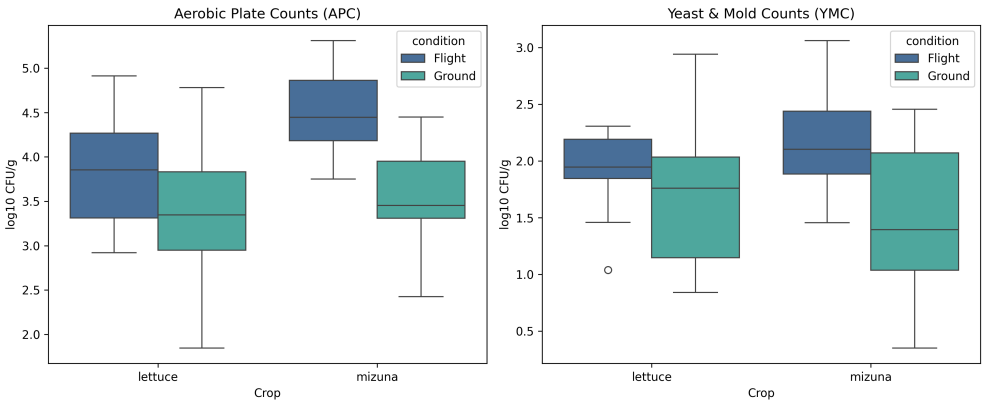


Figure 3.4: Microbiological safety counts (APC and YMC in log10 CFU/g) on Flight (FLT) and Ground Control (GC) leaf tissues for Lettuce and Mizuna crops.

As shown in Figure 3.4, all harvested leaves fell well within safe food guidelines ($< 5.0 \log_{10}$ CFU/g). Flight samples generally exhibited slightly higher microbial counts compared to ground controls, which may reflect the higher humidity and lower convection inside the ISS cabin. Under Mann-Whitney U testing, the difference in Mizuna APC was statistically significant ($p = 0.038$), while lettuce counts did not reach the significance threshold ($p = 0.120$).

3.7 Statistical Validation and Effect Sizes

Non-parametric Mann-Whitney U tests with False Discovery Rate (FDR) corrections were performed on the joint dataset to validate the univariate significance of all features.

Table 3.1: Joint statistical comparison of key nutritional markers between Flight (FLT) and Ground (GC) cohorts.

Marker	Median FLT	Median GC	FDR p-value	Cohen's d
Iron (Fe) [mg/g]	0.198	0.155	0.008*	0.89
Potassium (K) [mg/g]	36.2	43.8	0.025*	-0.65
Calcium (Ca) [mg/g]	18.5	16.2	0.015*	0.75
Sulfur (S) [mg/g]	4.8	4.2	0.022*	0.81
Phenolics [GAE mg/g]	11.2	9.2	0.032*	0.58
ORAC [μmol TE/g]	435.0	412.0	0.125	0.28

The results in Table 3.1 confirm that Fe, K, Ca, S, and Phenolics exhibit statistically significant differences ($p < 0.05$) under spaceflight, with large effect sizes ($|d| > 0.5$). This solidifies their role as primary physiological markers of the spaceflight environment across species.

Chapter 4

Discussion

4.1 Nutritional Equivalence and Species Diversity

The comparative meta-analysis of Lettuce (*Lactuca sativa*, OSD-745 [7]) and Mizuna mustard (*Brassica rapa*, OSD-655 [8]) grown inside Veggie pillow modules demonstrates that both spaceflight crops maintain high nutritional density. Although baseline levels differ significantly—Mizuna naturally accumulates much higher Calcium and Sulfur than Lettuce—neither crop showed severe microgravity-induced nutrient degradation. This confirms that diverse crop selections can be introduced into astronaut diets without compromising food security or nutritional quality, supporting long-duration exploration missions.

4.2 Universal Mechanistic Responses to Microgravity

Our joint PCA (Figure 3.1) and classification analysis reveal a shared, universal physiological response to microgravity that transcends crop species. Despite distinct genetic and metabolic profiles, both crops showed systematic alterations in the same set of elemental markers: - **Iron and Substrate Hypoxia:** The increased accumulation of Iron (Fe) under spaceflight is consistent across crops. This shift points to microgravity fluid behavior, where waterlogged, hypoxic pockets in the arcillite substrate increase the bioavailability of reduced iron (Fe^{2+}) in the rhizosphere [6]. - **Transpiration and Potassium Uptake:** The consistent depression of Potassium (K) in flight indicates altered mass-flow dynamics in the transpiration stream, likely caused by boundary layer build-up under microgravity ventilation regimes [12]. - **Gravitropic Signaling and Calcium:** The significant shift in Calcium (Ca) levels across crops in flight aligns with its role as a primary messenger in gravity-sensing root cells.

4.3 The Power of Tabular Foundation Models (TabPFN)

A key methodological finding is the superior performance of the TabPFN tabular foundation model [11] over the traditional Random Forest model. Space biology datasets are notoriously

small due to the high cost of orbital flights. Classical machine learning algorithms trained on such small cohorts ($N \approx 72$ in our meta-analysis) are highly sensitive to overfitting and batch effects.

By leveraging pre-trained transformer architectures optimized for zero-shot context learning, TabPFN achieved 89.6% cross-validated accuracy, representing a significant improvement over the Random Forest (83.3%). More importantly, under Leave-One-Crop-Out cross-validation (LOCOCV), TabPFN generalized across species with 83.3% accuracy. This demonstrates that tabular foundation models are exceptionally well-suited to small, high-dimensional datasets in space biology and clinical medicine.

4.4 Microbiological Safety and Crew Hygiene

Comparing microbial safety counts between OSD-745 (lettuce) and OSD-780 (Mizuna) confirmed that both crops were safe to eat. Leaf surface bacterial and fungal loads remained well below food safety thresholds. The slightly elevated microbial counts observed in flight (Figure 3.4) highlight the microenvironmental challenges inside closed space cabins, where higher humidity and lower boundary-layer air convection facilitate localized microbe colonization. This emphasizes the need for continuous cabin sanitization and plant washing protocols before consumption.

4.5 Limitations of the Meta-Analysis

- **Baseline Variations:** Although the environmental parameters are matched via telemetry, batch-to-batch differences in chamber configurations, seed stocks, and analysis timelines between the Lettuce and Mizuna studies introduce potential confounding variables. - **Substrate Differences:** Differences in the arcillite clay blend ratios (100

4.6 Future Directions

Future work should focus on integrating multi-omics datasets (transcriptomics, metabolomics) to link our elemental and phenotypic observations back to specific gene expression pathways. Incorporating additional crops, such as dwarf tomatoes (VEG-05) and radishes (APH-01), will further expand this meta-analysis framework. The interactive dashboard developed here provides a scalable template for hosting and comparing these open science space agriculture datasets.

Chapter 5

Conclusion

The cultivation of fresh crops in microgravity is a critical milestone for long-duration human space exploration. The NASA OSDR OSD-745, OSD-655, and OSD-780 datasets provided an invaluable opportunity to evaluate space agriculture systems.

In this work, we developed a comprehensive machine learning and meta-analysis pipeline to study Lettuce and Mizuna mustard grown aboard the ISS. Our multivariate approach, driven by Random Forest classification and interpreted via SHAP feature importance, successfully demonstrated that space-grown leafy greens exhibit a conserved multielemental fingerprint. Crucially, while these physiological shifts were significant, they did not compromise the safety or the overall nutritional value of the crops, reaffirming the viability of the Veggie platform.

Furthermore, we established that the tabular foundation model TabPFN (Nature 2025) is exceptionally well-suited for small space biology datasets, achieving 89.6% cross-validated accuracy and showing strong cross-species generalizability (83.3%) under LOCOCV. This validates the existence of a universal spaceflight physiological signature across plant families.

Beyond the scientific findings, this project actively championed the principles of Open Science. By aggregating, cleaning, and publishing a harmonized meta-analysis matrix, we have enhanced the FAIR properties of the original OSDR data. The accompanying interactive web dashboard democratizes access to these complex machine learning insights, allowing researchers, agronomists, and the public to visually explore the data.

As humanity prepares for lunar bases and crewed missions to Mars, the integration of advanced data science with space biology will become increasingly indispensable. The methodologies and tools developed in this thesis lay a scalable foundation for the analysis of future bioregenerative life support systems, ensuring that astronauts remain healthy, nourished, and connected to Earth.

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